

and servers) but now we are merging the two versions of the VM. In fact, VM is one of Oracle's crown jewels. You will see a lot of significant developments in JDK and Java SE for sure."

"The other component is the client (end user) side of Java. The reason why the client is very vital is that we are currently compared with technologies like .NET and others, simply because we have both the client as well as server side to our Java applications and SDKs. We are not like PHP (a scripting language originally designed for Web development) and others, where the focus is only on the server side. Even though we are not a client-focused consumer space company, I think Java's success has been in driving enterprise applications. Therefore, every investment you see on the client side will be more in tune with the enterprises. Our Java Micro Edition or Java ME (a Java platform designed for embedded systems, such as mobile devices) is a revenue-making business. Though there are contenders in the smartphone space, we feel there is a huge market for feature phones."

"Going forward, we are very clear that we are going to invest in Java Card (a technology that allows Java-based applications (applets) to be run securely on smart cards and similar small memory footprint devices), Java ME, the core JDK platform, Java FX and Java EE," she concluded.

What makes Java the apple of a developer's eye?

One might wonder why Java has become invaluable to software developers. In an earlier presentation, Steven G Harris, senior vice-president, Application Server Development, Oracle, who oversees product development for Java EE, said, "One of the great things about Java is that if you learn the language, the platform and the libraries, you have an incredible variety of deployment options and ways you can target the things you are developing—whether for the enterprise, desktop, or the mobile and embedded space."

According to the Tiobe Community Programming Community Index, Java remains the world's most popular programming language. Though Java has substantial distribution through the company, its language and platform work best on Oracle's middleware foundation. "All of our middleware—when we talk about Fusion Middleware—is written in Java," Harris noted. "So here's the message: If Java is not successful, we will fail. We really want Java to be successful, so keeping Java open, and having developers working in Java is really critical to our success."

Paving the way for Java's future development

Harris pointed to the estimated 14 million Java developers across the world with some nine million using Java regularly during the work day. He acknowledged that Java has been criticised as advancing too slowly and for overly cumbersome and bureaucratic processes. But he insisted that, "We are very committed to accelerating it and making Java more competitive, making it the kind of language you choose to develop with because of the advantages it gives you."

Oracle believes that its tie-ups with different companies

like IBM are going to reinforce Java tremendously. Earlier, it was fragmented with everybody creating their own flavours. But that should be a thing of the past, with Oracle reaching out to a lot of companies to come and participate in one space in order to grow Java together as a community. This was the vision for Java at Sun as well, Ramani quipped.

During the conference, the company also announced its plans for advancing the Java SE and optimising it for new application models and hardware, including extended support for scripting languages, increased developer productivity and lower operational costs.

The announced roadmap for the OpenJDK accelerates the availability of Java SE with two releases, one slated for 2011 and the other for 2012. These OpenJDK releases will continue to serve as the basis for the Oracle JDK7 and JDK 8.

Going forward...

Oracle is currently working to merge the Oracle Java HotSpot JVM and the Oracle JRockit JVM into a converged offering, which is aimed at leveraging the best features of each of these implementations. The company also plans to contribute the results of the combined Oracle Java HotSpot and Oracle JRockit JVMs to the OpenJDK project. The Oracle JDK and Java Runtime Environment or JRE (a software platform that contains JVM, Java libraries and some other components) will continue to be available as free downloads, with no changes to the existing licensing models.

It also outlined plans to deliver advanced graphics, high-fidelity media, and the ability to render HTML content within Java applications through a programming model that combines the power of Java with the ease of JavaFX. The company said it is working to tightly integrate JavaFX with the JRE to enable better performance and improved usability.

As part of the next release of JavaFX in the third quarter of 2011, Oracle will introduce a new set of Java APIs (application programming interfaces) that will open JavaFX capabilities to all Java developers, without the need for them to learn a new scripting language. The new Java APIs will allow the use of a host of Java features such as generics, annotations and multi-threading, and will also make it easier for Web developers to use JavaFX within other popular dynamic languages such as Ruby, Groovy and JavaScript.

The company is also mulling over other enhancements, including new language features, small-footprint CPU-efficient capabilities for cards, phones and TVs.

As conveyed during the conference, Oracle's business strategy and vision of Java's future would be to deliver a complete, open, integrated stack of hardware, infrastructure, database, middleware and business applications. The US software maker aims to make hardware and software work together to provide better serviceability and manageability. 

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